

Transaction Failure Analysis

Identifying patterns associated with failed transactions to support transaction reliability and operational improvement.

Failure Rate

5.25%

Total Transactions

12K

Success Rate

90.47%

Risk Review Rate

19.60%

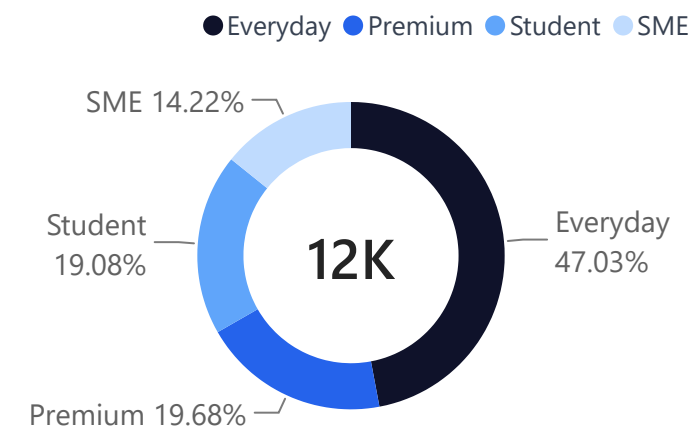
Executive Summary

- **12,000 transactions** were analysed across transaction, channel, device, customer, and risk-related attributes.
- **630 transactions failed**, representing a **5.25% failure rate**.
- **Risk-reviewed transactions** recorded a higher failure rate (**7.23%**) compared with non-risk-reviewed transactions (**4.77%**).
- Failure rates varied across **time of day and device type**, with **risk-reviewed iOS transactions** showing a higher failure rate (**9.26%**).
- Most customer and transaction characteristics showed **relatively similar failure rates**, with the more notable differences concentrated in specific transaction conditions rather than across the broader customer population.

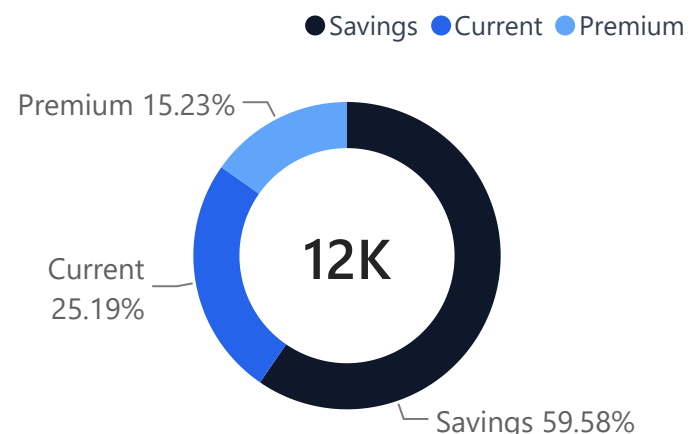
Transaction Profile

Overview of the customer, account, channel, device, and transaction characteristics of the analysed transaction population.

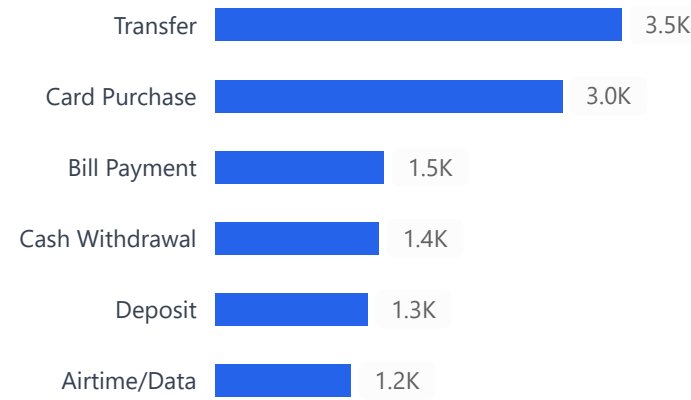
Customer Segment Distribution



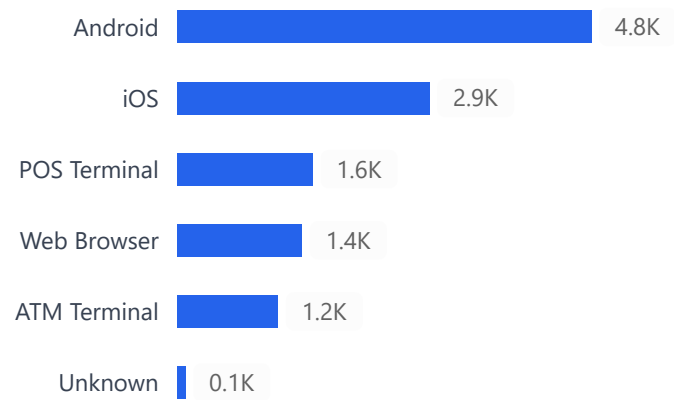
Account Type Distribution



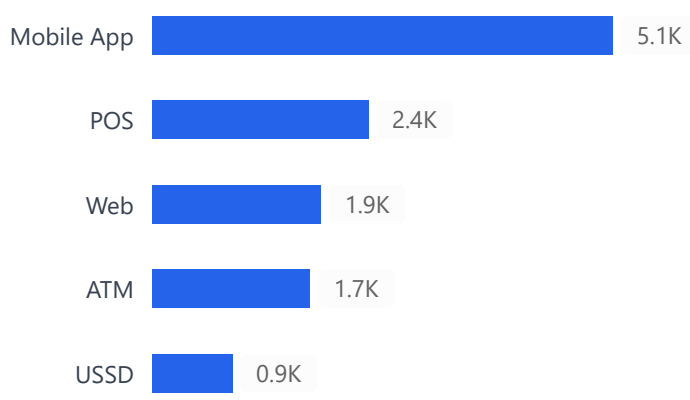
Transaction Type Distribution



Device Type Distribution



Preferred Banking Channel



Profile Summary

- The analysis covers **12,000 transactions** across multiple customer segments, account types, channels, and device types.
- Transaction activity was distributed across a diverse mix of customer segments, with **Everyday and Premium customers representing substantial shares** of the transaction population.
- Digital banking channels and mobile devices accounted for a significant proportion** of transaction activity, reflecting the range of electronic banking activity captured in the dataset.
- A broad range of transaction types was observed, providing a **diverse basis for assessing transaction failure patterns**.

Key Findings

Analysis of factors associated with transaction failures across the transaction population.

Failure Rate by Risk Review



Failure Rate by Device Type



Device_Type

All



Time_of_Day

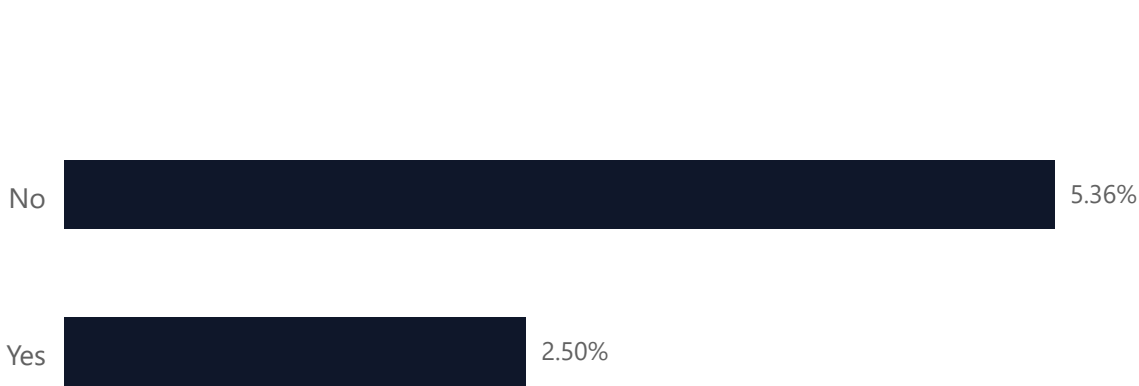
All



Failure Rate by Time of Day



Failure Rate by International Transaction Status



Key Insights

- Risk-reviewed transactions recorded a higher failure rate (**7.23%**) than non-risk-reviewed transactions (**4.77%**).
- **iOS transactions recorded the highest failure rate among device categories at 5.75%**, although the difference across device types was relatively modest.
- Failure rates were highest during the **Afternoon (5.93%)** and **Evening (5.90%)** periods.
- **Domestic transactions recorded a higher failure rate (5.36%) than international transactions (2.50%),** although international transactions represented a much smaller share of the transaction population.

Recommendations

Recommended actions to reduce transaction failures and improve transaction reliability.

Recommendation 1: Review Risk-Review Processes

Observed Pattern

Risk-reviewed transactions recorded a 7.23% failure rate, compared with 4.77% for non-risk-reviewed transactions.

Recommendation

Review risk rules, authentication requirements, and transaction processing controls to identify factors associated with the elevated failure rate.

Expected Impact

Improved transaction completion rates and more effective risk controls.

Recommendation 2: Investigate iOS Transactions

Observed Pattern

iOS transactions recorded the highest failure rate among device categories (5.75%). Within the risk-reviewed population, the iOS failure rate increased to 9.26%.

Recommendation

Review iOS transaction failures, focusing on authentication outcomes, risk-rule triggers, and transaction processing errors to identify potential device-specific issues.

Expected Impact

Improved transaction reliability, reduced transaction failures, and identification of potential device-specific technical or operational issues.

Recommendation 3: Monitor Peak Transaction Periods

Observed Pattern

Failure rates were higher during the Afternoon (5.93%) and Evening (5.90%) compared with the overall failure rate of 5.25%.

Recommendation

Monitor transaction performance during Afternoon and Evening periods, focusing on transaction volumes, system performance, processing capacity, and transaction error trends.

Expected Impact

Improved transaction reliability during higher-activity periods, reduced transaction failures, and enhanced customer experience.